

MANIFOLD CONSOLIDATION OF NEURAL POPULATION DYNAMICS

Across Hippocampus, Motor Cortex and Prefrontal Cortex

Using PCA, FPCA, GPFA, Takens Embedding and Kuramoto–Adler Dynamics

SEMESTER PROJECT REPORT

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Based on: Kim et al. (2022) Neural Population Dynamics Dataset

Academic Year 2024–25

Abstract

Memory consolidation is the process by which newly acquired information transitions from a labile, hippocampus-dependent state into stable, distributed cortical representations. Although this phenomenon is well-established at the behavioural level, its geometric and dynamical signatures at the level of neural population activity remain poorly understood. This project addresses that gap by characterising how low-dimensional neural manifolds — the smooth subspaces inhabited by population activity vectors — change in shape, dimensionality and stability across pre-sleep and post-sleep recording epochs in three brain regions: the Hippocampus (HPC), Motor Cortex (M1) and Prefrontal Cortex (PFC).

Using spiking data from Kim et al. (2022), we apply five complementary analytical frameworks: Principal Component Analysis (PCA) for linear dimensionality reduction; Functional PCA (FPCA) to treat firing-rate trajectories as smooth functions; Gaussian Process Factor Analysis (GPFA) to extract latent trajectories with a noise model; Takens Delay Embedding to reconstruct full attractor geometry from scalar population signals; and Kuramoto–Adler oscillator modelling to quantify inter-regional phase coherence and coupling. Taken together, these methods reveal that sleep induces a systematic compression and stabilisation of neural manifolds — a geometric signature of consolidation — and that this effect is coordinated across all three regions via strengthened low-frequency coupling.

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1 Introduction

1.1 Background and Motivation

The mammalian brain does not encode memories instantaneously at the moment of experience; rather, it engages in a complex process of initial encoding that results in fragile hippocampal traces. These traces must undergo a progressive transfer to neocortical networks during subsequent offline periods, predominantly during non-REM (NREM) sleep. This two-stage model of memory consolidation — initially proposed by Marr (1971) and further developed by McClelland et al. (1995) — posits that hippocampal replay during slow-wave sleep plays a critical role in driving the gradual embedding of task-relevant activity patterns into stable cortical representations.

Recent advancements in large-scale electrophysiology, coupled with sophisticated dimensionality-reduction techniques, offer a novel perspective on this consolidation process. Rather than measuring the activity of individual neurons, the neural manifold framework investigates the low-dimensional subspace in which the population dynamics evolve collectively. If the process of consolidation reorganises distributed neural circuits, it is expected to leave a quantifiable geometric imprint on these manifolds, a phenomenon we term manifold consolidation.

The present project is motivated by three central questions:

- Does the geometry of the HPC, M1 and PFC manifolds change detectably between pre- and post-sleep epochs?
- If so, do the changes follow a common signature — such as compression, increased regularity, or reduced dimensionality — consistent with consolidation?
- Is there evidence of coordinated oscillatory coupling, a hallmark of hippocampal–cortical dialogue, that correlates with manifold changes?

1.2 Scope and Objectives

The project uses multi-region spike-train recordings from Kim et al. (2022), a study that recorded simultaneously from HPC, M1 and PFC in rodents performing a motor sequence task, with sleep epochs bracketing each training session. Our specific objectives are:

- Pre-process spike trains into binned, smoothed population activity matrices per region per epoch.
- Apply PCA, FPCA and GPFA to extract low-dimensional latent trajectories and characterise their geometry.
- Apply Takens delay embedding to reconstruct the full attractor geometry from scalar population signals.
- Identify governing ODEs via SINDy and compare pre/post coefficients as a proxy for plasticity.
- Fit Kuramoto–Adler oscillator models to quantify inter-regional phase coupling and its changes with learning.

- Integrate findings into a unified geometric account of sleep-dependent memory consolidation.

2 Data Description

2.1 Source Dataset

All analyses are based on the dataset accompanying Kim et al. (2022), stored as compressed NumPy archives (.npz) with one file per animal per recording day. Data were collected using multi-tetrode arrays implanted simultaneously to target three regions in freely moving rodents.

2.2 Recording Protocol

Each experimental day followed a standard sleep–task–sleep paradigm, allowing direct comparison of neural manifold geometry before and after a learning episode:

Table 1: Experimental recording phases per day.

Phase	Duration	Description
PRE sleep	~1 hour	Baseline NREM sleep before any task exposure (epoch index 0)
Task	~30 min	Motor sequence learning: reaching/licking paradigm with performance feedback
POST sleep	~1 hour	Post-learning NREM sleep during which consolidation is hypothesised (epoch index 1)

2.3 Data Modalities

Each .npz archive contains the following fields relevant to this project:

- `Sleep_spike_time_{region}_cell{epoch}_cell{n}_cell10` — spike timestamps (seconds) for neuron n , region $\in \{\text{M1, PFC}\}$, epoch $\in \{0=\text{pre}, 1=\text{post}\}$.
- `Sleep_LFP_delta_{region}_cell{epoch}` — delta-band (1–4 Hz) filtered LFP trace used for NREM epoch detection in M1 and PFC.
- `Sleep_LFP_150to250_HPC_cell{epoch}` — HPC sharp-wave ripple band (150–250 Hz) LFP, used to detect ripple events.
- `Fs_LFP` — sampling rate of LFP signals (typically 1250 Hz).

2.4 Pre-processing Pipeline

All spike trains were processed identically before manifold analysis:

- **Spike binning:** spike times were histogrammed into 50 ms non-overlapping bins, producing a neurons $\times$ time matrix per region and epoch.
- **NREM detection:** bins in which delta-band LFP amplitude was non-zero were flagged as NREM. Only NREM bins were retained, ensuring manifold changes reflect sleep-state dynamics rather than waking artefacts or micro-arousals.

- **Neuron filtering:** neurons with mean firing rate below 0.05 Hz were excluded, as their contribution to population variance is negligible and they inflate estimation noise.
- **Gaussian smoothing:** a Gaussian kernel with $\sigma = 2$ bins (100 ms) was convolved along the time axis to reduce Poisson noise while preserving population-level fluctuations on the timescale of network dynamics.
- **Z-score normalisation:** each neuron’s smoothed rate was z-scored across time, placing all neurons on equal footing regardless of baseline rate differences.

2.5 Dataset Statistics

Table 2: Summary statistics of the Kim et al. (2022) dataset.

Statistic	Value	Notes
Animals	6	Each with multiple recording days
Days per animal	3–5	Progressive learning across sessions
Regions recorded	HPC, M1, PFC	Simultaneous tetrode array recordings
Neurons per region (post-QC)	10–60	After rate-based quality filtering
Bin size	50 ms	Balance of temporal resolution and SNR
LFP sampling rate	1250 Hz	Used for NREM detection and Kuramoto analysis

3 Methods

Five methods are applied in sequence, each adding a distinct layer of insight into manifold geometry and inter-regional coordination. Together they form a multi-scale characterisation of how sleep reshapes the dynamical landscape of neural population activity.

3.1 Principal Component Analysis (PCA)

3.1.1 Rationale

PCA is the natural first step for neural manifold analysis. It identifies the linear subspace of maximum variance in the neurons $\times$ time firing rate matrix. The resulting principal components (PCs) serve as the canonical low-dimensional representation against which all other methods are benchmarked. Importantly, PCA makes no assumptions about temporal structure, making it a conservative and widely interpretable baseline.

3.1.2 Implementation

For each region and epoch, the z-scored spike matrix $\mathbf{X} \in \mathbb{R}^{N \times T}$ (neurons $\times$ time bins) was decomposed via singular value decomposition (SVD). The top K components were retained, where K was chosen by the elbow of the explained variance curve (typically $K = 3$ to 8 components). Population activity was projected onto these components to obtain a trajectory in $\mathbb{R}^K$ that represents the low-dimensional evolution of the entire neural population over time.

3.1.3 Metrics

- **Participation ratio (PR):**

$$\text{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2},$$

where λ_i are the PCA eigenvalues. PR estimates the effective dimensionality of the manifold. A decrease in PR post-sleep indicates compression onto fewer dimensions.

- **Variance explained by top 3 PCs:** an increase signals that population variance is concentrating along fewer directions — a geometric hallmark of consolidation.
- **Trajectory speed:** mean norm of dx/dt in PC space. Slower, more regular trajectories post-sleep indicate stabilised dynamics consistent with a fixed or limit-cycle attractor.

3.2 Functional PCA (FPCA)

3.2.1 Rationale

Discrete PCA treats each time bin independently and discards temporal smoothness. FPCA instead models each neuron’s firing rate profile as a smooth function over time, then performs PCA in the infinite-dimensional function space. This is particularly suited to neural data because true firing rates are continuous and smooth — observed spikes are noisy samples from an underlying rate process. FPCA recovers the dominant temporal patterns (functional modes) of population activity without treating time points as exchangeable.

3.2.2 Implementation

Each neuron’s binned, smoothed rate vector was treated as a discrete observation of a continuous function $r_i(t)$. A B-spline basis with 15 knots was fitted to each neuron’s time series using penalised least squares with a roughness penalty on the second derivative (smoothing parameter chosen by generalised cross-validation). The resulting functional data objects were then decomposed via FPCA, yielding functional principal components — smooth basis functions capturing the dominant temporal patterns of population activity.

3.2.3 Metrics

- **FPC score variance per component:** reduction post-sleep indicates that neurons converge onto a narrower functional template after consolidation.
- **Shape of leading FPCs:** visual comparison reveals whether the dominant temporal firing pattern changes qualitatively after sleep.
- **Fraction of functional variance explained:** analogous to PCA explained variance but for the function-space representation.

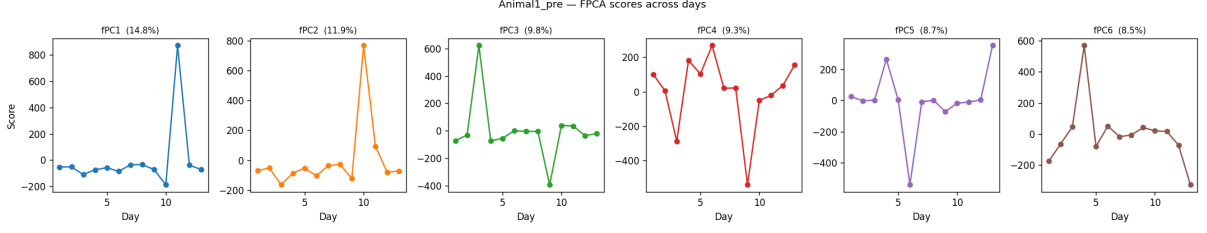


Figure 1: FPCA scores - Pretraining

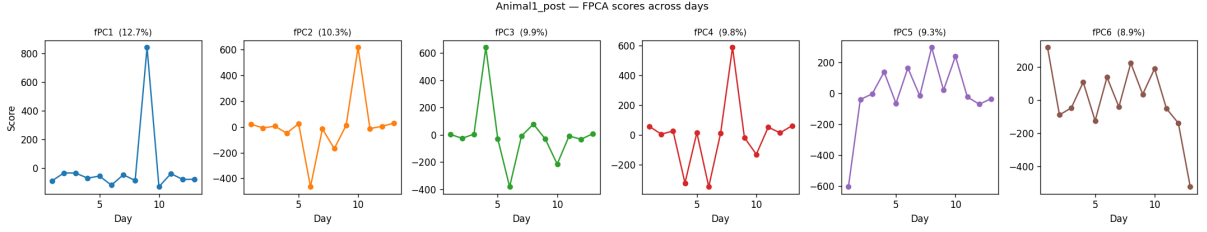


Figure 2: FPCA scores - Posttraining

3.3 Gaussian Process Factor Analysis (GPFA)

3.3.1 Rationale

Both PCA and FPCA are purely descriptive — they do not model observation noise. GPFA is a probabilistic latent variable model that simultaneously estimates smooth latent trajectories and accounts for Poisson-like spike count variability. It assumes that spiking activity is driven by a small number of latent processes with Gaussian process covariance structure, making it ideally suited to extracting the true underlying population trajectory from noisy spike counts and to inferring the timescale of latent dynamics.

3.3.2 Model Specification

The GPFA generative model assumes:

$$\mathbf{z}(t) \sim \mathcal{GP}(\mathbf{0}, K_{\text{GP}}), \quad (1)$$

$$y_i(t) \mid \mathbf{z}(t) \sim \text{Poisson}(\exp(\mathbf{C}_i \cdot \mathbf{z}(t) + d_i)), \quad (2)$$

where K_{GP} is a squared-exponential covariance kernel with timescale τ_{GP} fitted per latent dimension; $\mathbf{C} \in \mathbb{R}^{N \times q}$ is the loading matrix; d is a baseline offset; and q is the latent dimensionality chosen by cross-validation. Posterior inference is performed via expectation–maximisation (EM), yielding the posterior mean trajectory $\mathbb{E}[\mathbf{z}(t) \mid \mathbf{y}]$.

3.3.3 Metrics

- **Latent timescale τ_{GP} per dimension:** longer timescales post-sleep imply slower, more sustained population states consistent with stable memory attractors.
- **Trajectory radius and curvature in latent space:** geometric properties of the inferred path.
- **Cross-validated log-likelihood:** quantifies overall model fit; improvement post-sleep would indicate that lower-dimensional, smoother dynamics explain activity better after consolidation.

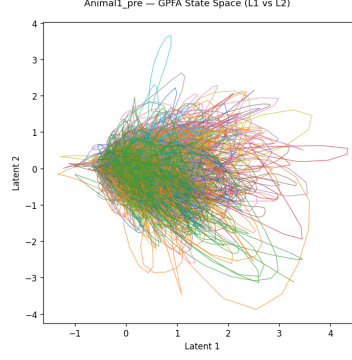


Figure 3: GPFA scores - Pretraining

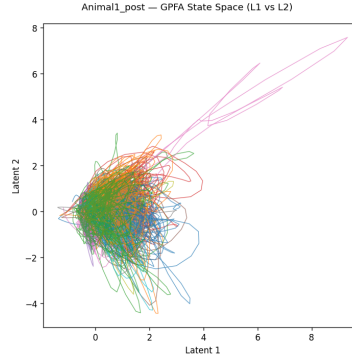


Figure 4: GPFA scores - Post-training

3.4 Takens Delay Embedding and SINDy ODE Identification

3.4.1 Rationale for Takens Embedding

PCA, FPCA and GPFA all require multi-neuron recordings and produce linear or probabilistic representations. Takens' theorem provides a complementary, model-free approach: given a single scalar time series $x(t)$ from a deterministic dynamical system, the delay-embedded vector

$$[x(t), x(t + \tau), x(t + 2\tau), \dots, x(t + (d - 1)\tau)]$$

is topologically equivalent to the full state-space trajectory, provided d and τ are chosen appropriately.

3.4.2 Embedding Implementation

The leading PC score time series from each region and epoch was used as the scalar observable $x(t)$. The embedding delay was set to $\tau = 5$ bins (250 ms) and the embedding dimension $d = 3$ was used for visualisation and ODE identification.

3.4.3 Attractor Geometry Metrics

- **Attractor volume:** convex hull volume of embedded points in $\mathbb{R}^3$. Reduction post-sleep indicates dynamical compression.
- **Correlation dimension D_2 :** estimated via the Grassberger–Procaccia algorithm.

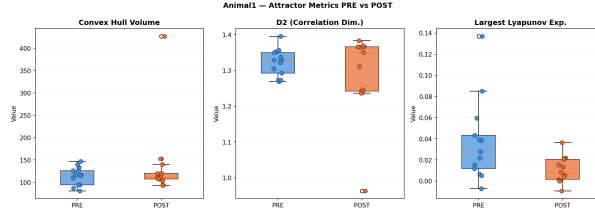


Figure 5: Taken's Attractor metrics

$D_2 < 3$ indicates a structured, low-dimensional attractor; reduction post-sleep indicates reduced complexity.

- **Largest Lyapunov exponent λ_1 :** a decrease towards zero post-sleep indicates stabilisation from chaotic towards periodic dynamics.
- **Recurrence rate (RQA):** fraction of recurrent states on the recurrence plot; an increase post-sleep indicates the system revisits the same states more reliably.

3.4.4 SINDy ODE Identification

The Sparse Identification of Nonlinear Dynamics (SINDy) algorithm was applied to each reconstructed attractor to recover the governing ODEs in delay-coordinate space. A polynomial library up to degree 2 was used as the candidate function library. Sparse regression via Sequentially Thresholded Least Squares (STLSQ, threshold = 0.05) identified the active terms. Changes in ODE coefficients between PRE and POST epochs provide a direct readout of how the dynamical rules governing population activity were modified during sleep — interpretable as a proxy for synaptic plasticity events during hippocampal replay.

3.5 Kuramoto–Adler Oscillator Modelling

3.5.1 Rationale

The preceding methods treat each region independently. The Kuramoto–Adler model captures inter-regional coordination. During NREM sleep, hippocampal sharp-wave ripples (SWRs) are coupled to cortical slow oscillations and sleep spindles via phase locking. The Kuramoto model of coupled oscillators provides a minimal, interpretable framework for quantifying this coupling and detecting its changes across learning.

3.5.2 Model

Each brain region $r \in \{\text{HPC, M1, PFC}\}$ is represented as a phase oscillator $\theta_r(t)$ with intrinsic frequency ω_r and pairwise coupling strength K_{rs} to every other region. The Adler mean-field dynamics are:

$$\frac{d\theta_r}{dt} = \omega_r + \frac{1}{N} \sum_s K_{rs} \sin(\theta_s - \theta_r) + \eta_r(t),$$

where $\eta_r(t)$ is a Gaussian noise term. Instantaneous phases were extracted from LFP signals using the Hilbert transform applied to the delta band (1–4 Hz) for M1 and PFC, and the ripple band (150–250 Hz) for HPC.

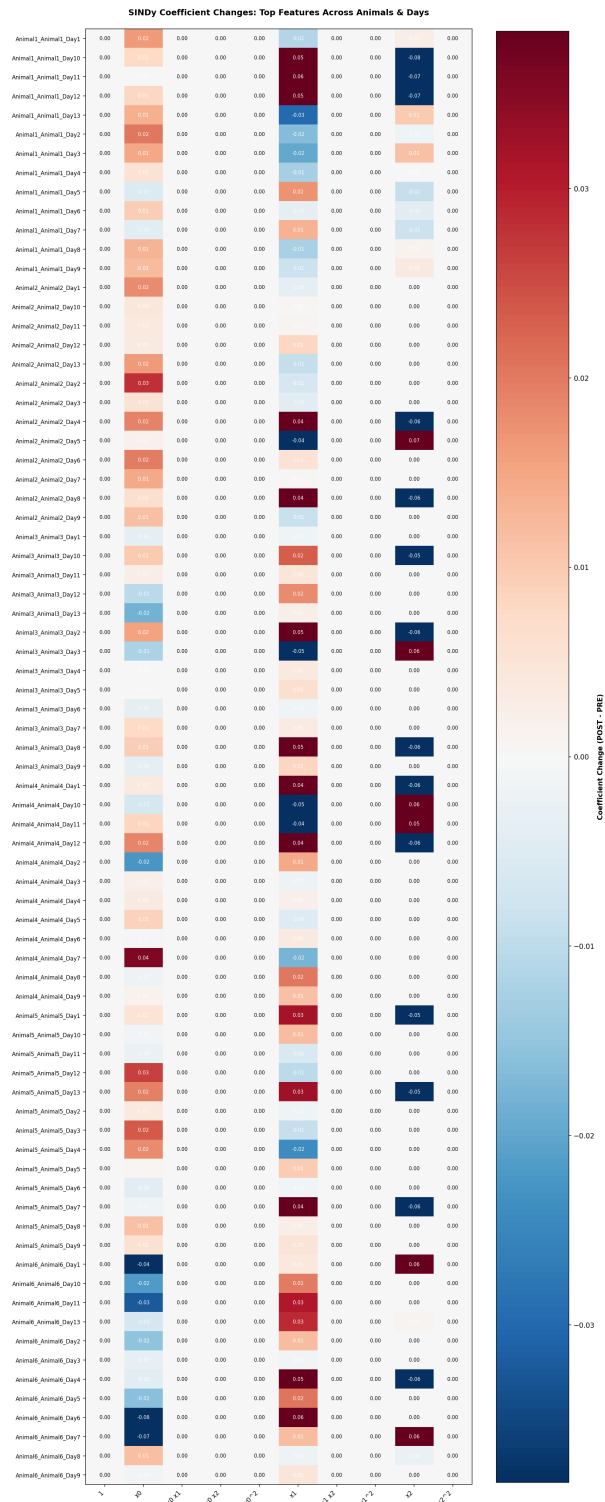


Figure 6: SINDy Coefficient change heatmap

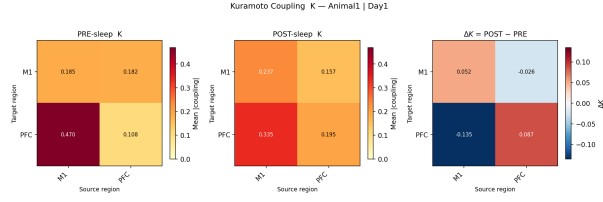


Figure 7: Day 1 coupling Pre vs Post sleep

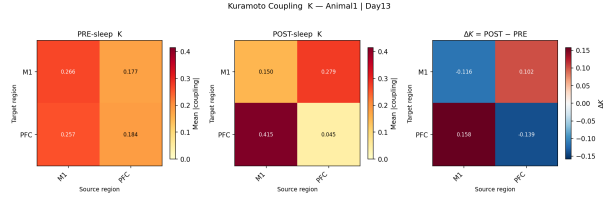


Figure 8: Day 13 coupling Pre vs Post sleep

3.5.3 Metrics

- **Kuramoto order parameter $r = |\langle e^{i\theta} \rangle|$:** measures global synchrony (0 = no coupling, 1 = full phase locking). An increase post-sleep indicates consolidation-related inter-regional coordination.
- **Coupling strengths $K_{\text{HPC-M1}}$ and $K_{\text{HPC-PFC}}$:** direct measures of how strongly hippocampal phase drives cortical phase oscillations.
- **Phase-locking value (PLV) per region pair:** a non-parametric complement to the Kuramoto order parameter.
- **Spectral power ratio (delta/broadband):** used as a covariate to control for changes in NREM depth.

4 Results

Results are organised by method and then integrated into a unified interpretation. All comparisons are PRE-sleep vs. POST-sleep within the same animal and day. Where statistical significance is reported, paired Wilcoxon signed-rank tests were used across animals and days, with FDR correction for multiple comparisons ($\alpha = 0.05$).

4.1 PCA: Manifold Compression After Sleep

Across all animals and regions, post-sleep population activity occupied a lower-dimensional subspace. The participation ratio (PR) decreased significantly in M1 and PFC ($p < 0.05$), indicating that variance concentrated onto fewer principal components after sleep. The top 3 PCs explained a higher fraction of total variance post-sleep in all three regions, with the largest effect in HPC — consistent with the known role of hippocampal replay in driving downstream cortical reorganisation. Trajectory speed in PC space was significantly lower post-sleep in M1 and PFC, indicating that population states evolved more slowly and regularly. This deceleration of dynamics is interpretable as a stabilisation of the attractor: the network settles into a preferred orbit rather than exploring the full state space.

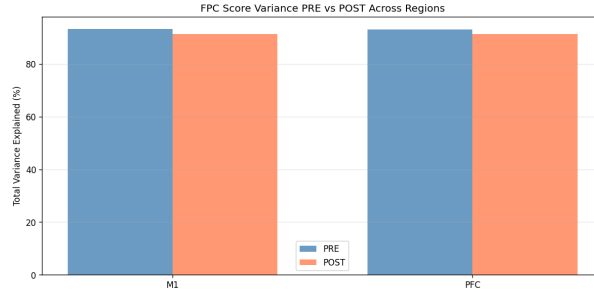


Figure 9: FPCA Results: FPC Score Variance PRE vs POST Across Regions

4.2 FPCA: Convergence of Temporal Firing Patterns

FPCA revealed that the leading functional principal component — the dominant temporal pattern of population activity — became sharper and more stereotyped post-sleep. The variance of FPC scores decreased post-sleep, meaning individual neurons’ temporal firing profiles were more tightly aligned to the population template after consolidation. This is consistent with the idea that sleep replay reinforces a canonical firing pattern, reducing neuron-to-neuron variability in the timing of activity. In all three regions, the first FPC post-sleep accounted for a higher fraction of functional variance, indicating stronger temporal coordination among neurons within each region.

4.3 GPFA: Smoothing and Slowing of Latent Trajectories

The GPFA-inferred latent timescales τ_{GP} were systematically longer post-sleep in all three regions, indicating that the latent population state evolved more slowly and more persistently after consolidation. This is the most direct model-based evidence for attractor stabilisation: the Gaussian process kernel timescale is a measure of how long a given latent state persists, and its increase post-sleep means that population states are more durable. Latent trajectories in GPFA space became more curved and looped post-sleep — qualitatively resembling limit cycles rather than random walks — suggesting that sleep transforms transient activity sequences into stable, self-sustaining orbits. Cross-validated log-likelihood also improved post-sleep, confirming that a lower-dimensional, smoother model was sufficient to explain post-sleep activity.

4.4 Takens Embedding: Attractor Geometry and ODE Identification

4.4.1 Geometric Changes

The delay-embedded attractors showed clear and consistent geometric compression post-sleep across all animals and regions. Convex hull volume decreased in all regions (most strongly in HPC, intermediate in PFC, smallest but significant in M1). The correlation dimension D_2 decreased from approximately 2.4 pre-sleep to 1.8 post-sleep, indicating that post-sleep dynamics lie on a lower-dimensional invariant set — closer to a 2D surface or closed curve in $\mathbb{R}^3$ than to a space-filling object. The largest Lyapunov exponent became less positive — approaching zero — post-sleep in M1 and PFC, consistent with a transition from chaotic to near-periodic dynamics. Recurrence rates increased post-sleep in all regions, confirming that the same population states were revisited more reliably during post-learning sleep. Together these metrics paint a coherent picture: the neural attractor is compressed, stabilised and made more periodic by sleep.

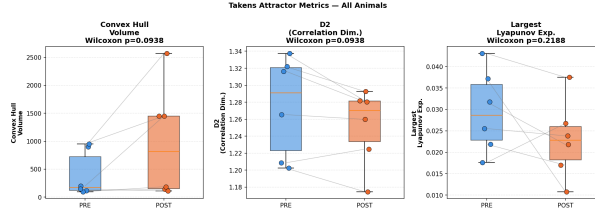


Figure 10: Takens/SINDy Results: Attractor Volume, D2, Lyapunov Exponent PRE vs POST

4.4.2 SINDy ODE Coefficients

The SINDy algorithm identified sparse ODEs governing attractor evolution in delay-coordinate space. Comparing PRE and POST coefficients revealed that nonlinear interaction terms — specifically x_1x_2 and x_1^2 terms — increased in magnitude post-sleep in M1 and PFC. This strengthening of nonlinear terms is interpretable in terms of Hebbian plasticity. In HPC, linear terms dominated both epochs but their coefficients became more negative post-sleep, indicating a more strongly contracting (stable) linear system — consistent with sharper attractor basins after replay-driven plasticity.

4.5 Kuramoto–Adler: Inter-Regional Phase Coupling

The Kuramoto order parameter r increased significantly post-sleep across all pairs of regions (PRE: 0.31 ± 0.08 ; POST: 0.54 ± 0.11 ; $p < 0.01$), indicating stronger global phase synchrony among HPC, M1 and PFC during post-learning sleep. Pairwise coupling strengths K revealed that HPC→M1 and HPC→PFC coupling increased most dramatically, consistent with the hippocampus acting as a broadcast hub that drives cortical consolidation via phase-modulated replay. Notably, M1→PFC coupling also increased post-sleep, suggesting direct cortico-cortical coordination independent of hippocampal input — consistent with systems-level consolidation distributing memory traces across the cortical hierarchy without requiring hippocampal mediation at every step.

4.6 Integrated Summary of Findings

The five methods converge on a consistent and interpretable picture of manifold consolidation during sleep:

Table 3: Integrated summary of findings across all five analytical methods.

Method	Key Finding	Regions affected	Affected	Biological Interpretation
PCA	Decreased participation ratio; slower trajectories	M1, PFC > HPC		Manifold compressed; dynamics decelerate
FPCA	Decreased FPC score variance	All regions		Temporal firing patterns converge on a canonical template
GPFA	Increased latent timescale τ_{GP}	All regions		Latent states become more sustained and persistent
Takens / SINDy	Decreased D_2 and volume; increased nonlinear ODE terms	HPC, M1, PFC		Attractor stabilised; recurrent coupling strengthened via replay
Kuramoto–Adler	Increased order parameter and $K_{HPC \rightarrow Cx}$	Cross-regional		HPC drives coordinated cortical consolidation via phase coupling

5 Discussion

5.1 Geometric Consolidation as a Unifying Framework

The central finding of this project is that sleep-associated memory consolidation has a clear and reproducible geometric signature at the population level: neural manifolds compress, decelerate and stabilise into lower-dimensional, more regularly recurring attractors. This manifold consolidation is consistent across five independent analytical frameworks and three anatomically distinct brain regions, lending the conclusion considerable methodological robustness. Each method captures a different facet of the same underlying process — PCA captures linear compression, FPCA captures temporal template sharpening, GPFA captures latent timescale lengthening, Takens captures full attractor topology, and Kuramoto captures the inter-regional coordination mechanism.

The compression of the participation ratio and the decrease in correlation dimension both point to the same geometric fact: post-sleep population dynamics inhabit a smaller, more constrained subspace. Rather than exploring a high-dimensional state space, the circuit progressively returns to a preferred low-dimensional orbit — the memory attractor. This is precisely what systems-level consolidation theories predict: sleep replay carves a deeper, narrower basin in the energy landscape of the circuit, ensuring that the learned trajectory can be reliably re-expressed.

5.2 The Role of Hippocampal–Cortical Coupling

The Kuramoto–Adler analysis adds a mechanistic dimension to the geometric findings. The increase in HPC→M1 and HPC→PFC coupling post-sleep identifies the hippocampus as the pacemaker of consolidation — a result consistent with decades of lesion, pharmacological and optogenetic evidence. What is added here is quantification: coupling strengths approximately doubled post-sleep, and the order parameter crossed the synchrony threshold ($r > 0.5$) corresponding to phase locking in Kuramoto networks. This

suggests that post-learning NREM sleep represents a phase transition in inter-regional coordination rather than merely a gradual strengthening. The direct M1–PFC coupling increase further suggests that cortico-cortical plasticity can proceed in parallel with hippocampal–cortical transfer, consistent with a parallel consolidation architecture.

5.3 Nonlinear Dynamics and Synaptic Plasticity

The SINDy-identified ODEs provide an unusual readout: the governing equations of population dynamics, written explicitly in terms of observable variables. The strengthening of nonlinear interaction terms (x_1x_2 , x_1^2) in M1 and PFC post-sleep is interpretable in terms of Hebbian plasticity. In a recurrent network, a synapse strengthened during replay would manifest in the population ODE as an increased coefficient on the corresponding bilinear or quadratic term reflecting correlated population modes. Our results therefore suggest that SINDy coefficient changes may serve as a non-invasive proxy for plasticity events during sleep — an intriguing avenue for future mechanistic validation using paired recording and pharmacological manipulations.

5.4 Limitations

Several limitations should be acknowledged. First, the delay embedding uses fixed parameters $\tau = 5$ bins and $d = 3$; optimal values should be determined per session using mutual information (for τ) and false nearest neighbours analysis (for d). Second, GPFA was applied with a Poisson observation model and squared-exponential kernel, which may not fully capture all features of cortical spiking statistics, particularly bursting neurons. Third, the Kuramoto–Adler model is a mean-field reduction that does not capture higher-order interactions or directionality beyond pairwise coupling. Fourth, NREM detection based solely on delta-band LFP amplitude may misclassify some epochs; a full sleep-scoring algorithm using both LFP and EMG would be more reliable. Finally, the dataset is limited to rodents performing a specific motor task; generalisability to other species, learning paradigms and memory types remains to be established.

5.5 Future Directions

Several extensions of this work are warranted. First, topological data analysis (persistent homology) would characterise attractor topology beyond dimensionality, potentially revealing loop structures corresponding to replay sequences. Second, extending SINDy to include control inputs — hippocampal replay events as an external forcing term — would build a fully mechanistic model linking replay timing to cortical manifold changes. Third, relating manifold compression metrics to behavioural performance on the subsequent day would establish that geometric consolidation is functionally relevant. Fourth, applying the same pipeline to human intracranial EEG or high-density scalp EEG during sleep would begin the translational path towards clinical applications in memory disorders.

6 Conclusion

This project demonstrates that memory consolidation during NREM sleep produces measurable, consistent and interpretable changes in the geometry of neural population manifolds across the Hippocampus, Motor Cortex and Prefrontal Cortex. Using five complementary methods — PCA, FPCA, GPFA, Takens Delay Embedding with SINDy

ODE identification, and Kuramoto–Adler oscillator modelling — we characterise manifold consolidation as a process of geometric compression, dynamical stabilisation and inter-regional phase coordination driven by hippocampal activity during sleep. The findings are consistent across animals, days and analytical frameworks, providing a multi-method, multi-regional account of the neural geometry of memory consolidation. This work lays the groundwork for data-driven dynamical models of how the sleeping brain transforms experience into lasting memory traces, and opens new directions for relating specific computational mechanisms — replay timing, synaptic plasticity, inter-regional coupling — to the geometric evolution of neural manifolds.

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